PROJECT PLAN / VERSION 1

Betting on Hong Kong's Weather, with Data

A machine learning plan to find mispriced temperature contracts on Polymarket

The idea in six points

Build a model that estimates the chance the day's top temperature reaches a cutoff, compare it with the Polymarket price, and trade only when the gap is big enough to pay for fees.

1 Linear benchmark

Fit a linear model first (Ridge/Lasso for the temperature, logistic regression for yes/no cutoffs). CART and XGBoost must beat it on walk-forward Brier score and log-loss.

2 Trading rule

Trade only when our probability beats the market's by enough to cover costs. Buy **Yes** if $P_{model} > P_{market}$ + fee + min edge. Buy **No** if $P_{model} < P_{market}$ - fee - min edge.

3 Taker fees

We are the taker, so the fee (up to 5 points, to be verified) is a cost on the market side. It is never added to our probability. Resting limit orders that get filled may avoid it.

4 Definitions

Instance, attribute, target variable, class, parameter, P_{model}, P_{market}, edge. Each defined in plain language on page 3.

5 Attributes

The input columns: past weather, rolling averages, sun and wind, the HKO forecast, and time of year. Market prices stay out of the model.

6 Who is affected

Traders, counterparties, liquidity providers, Polymarket, other traders, HKO and instructors. Key risks: settlement mismatch, access, slippage.

How the pieces fit. The linear model produces P_{model}. The trading rule compares it with P_{market} after the fee. If the model is well calibrated, the edge it finds is the expected profit per share. Calibration matters more than raw accuracy.

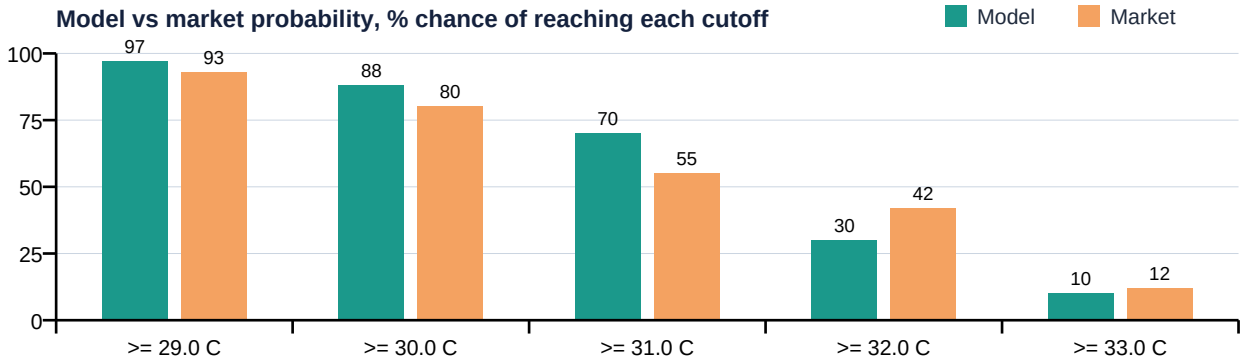
Trading rule and fees

We are the **taker**: we cross the spread and take an existing order, so the fee raises what we pay. Our probability stays an honest weather estimate.

Worked example (illustrative numbers, 5-point fee, 3-point minimum edge)

Contract	P_model (Yes)	P_market (Yes)	Yes edge	No edge	Decision	EV / share
Max temp >= 29.0 C	97%	93%	-1 pts	-9 pts	NO TRADE	-
Max temp >= 30.0 C	88%	80%	+3 pts	-13 pts	NO TRADE	-
Max temp >= 31.0 C	70%	55%	+10 pts	-20 pts	BUY YES	10 pts
Max temp >= 32.0 C	30%	42%	-17 pts	+7 pts	BUY NO	7 pts
Max temp >= 33.0 C	10%	12%	-7 pts	-3 pts	NO TRADE	-

Example probabilities are placeholders, not real market data.



The rule, step by step

Yes edge	P_{model} - (P_{market} + fee). Trade Yes only if this beats the minimum edge.
No edge	(1 - P_{model}) - ((1 - P_{market}) + fee). Trade No only if this beats the minimum edge.
Why not add the fee to the model?	That would make us trade more when fees mean we should trade less.
Check the cutoffs	Higher cutoffs are harder to reach, so probabilities must fall as the cutoff rises. A market that breaks this order shows an arbitrage flag.

To verify: the 5% fee figure, whether it is points per share or a percent of trade value, and whether these specific markets charge a taker fee at all.

Definitions

Instances are rows, attributes are inputs, the target variable is the output, and classes are the answer choices of a categorical target.

Term	Plain-language meaning	In this project
Instance	One row of the dataset	One calendar day
Attribute	A column the model reads as input	Past temperature, humidity, HKO forecast
Target variable (Y)	The column the model predicts	The day's max temperature, e.g. 31.4 C
Class	One possible value of a categorical target	0 or 1 per cutoff, or a bracket like 31-32 C
Parameter	A setting we choose, not a data column	Cutoff, number of cutoffs, fee, min edge
P_model	Our estimated chance the cutoff is reached	Model output
P_market	The Polymarket Yes price read as a probability	60 cents = 0.60
Edge / EV	Average profit per share after the fee	P_model - (P_market + fee)
Walk-forward validation	Train on the past, test on the next stretch, roll on	Prevents future leakage
Calibrated	A 70% call happens about 70% of the time	Reliability diagrams
Dutch book	Bracket prices sum to more or less than $1.00	Sum of 1.15 or 0.88

Attributes (model inputs)

Past weather	Max temp, min temp, mean humidity, sea-level pressure and rainfall on days t-1, t-2 and t-7
Engineered	3-day and 7-day moving averages of temperature and humidity; pressure change between t-1 and t-2
Sun and wind	Solar radiation, bright sunshine hours, wind direction, mean wind speed
Official forecast	HKO 9-day forecast max temperature, issued on t-1
Season	Sin_Day and Cos_Day, the day of year as a smooth cycle

Not an attribute: the actual max temperature is the target. Polymarket price, volume and liquidity are trading quantities, used at the trading step and kept out of the model so its probability stays independent of the crowd's. Your original brief lists them as attributes, so state the choice in the report.

To verify: the listed data sources clearly cover daily temperatures and the 9-day forecast. Humidity, pressure, rainfall, solar, sunshine and wind need their own HKO datasets.

Who is affected

Us (the traders)	Bear profit or loss, model risk and fees
Counterparties	Casual bettors and market makers lose when our model is right
Liquidity providers	May widen spreads or pull orders if we trade against them consistently
Polymarket	Earns our fees and benefits from volume; controls fee and settlement rules
Other traders	Prices get slightly more accurate with informed participants
HKO	Not financially affected; supplies the data and the reading that settles contracts
ISOM 3360 instructors	Grade modeling rigor, not trading profit

Risks to note

Settlement mismatch

Polymarket's resolution rule or rounding may differ from the HKO figure we trained on. At a 31.0 threshold, one decimal matters.

Access

Whether Polymarket access is permitted from Hong Kong has not been verified.

Liquidity and slippage

Thin order books can erase a small edge at larger trade sizes.

Next steps

Confirm the fee schedule and settlement rule, find datasets for the unconfirmed attributes, then build the linear benchmark and the walk-forward backtest with net profit after fees.

Verified public data sources

Source	Link
Smart lampposts real-time experimental meteorological data (post quality checks)	https://data.gov.hk/en-data/dataset/hk-hko-rss-smart-lamppost-weather-data
9-day weather forecast	https://data.gov.hk/en-data/dataset/hk-hko-rss-9-day-weather-forecast
Daily maximum, mean and minimum temperatures	https://data.gov.hk/en-data/dataset/hk-hko-rss-daily-temperature-info-hko
Daily Maximum Temperature All Year - Hong Kong Observatory	https://data.gov.hk/en-data/dataset/hk-hko-rss-daily-temperature-info-hko/resource/1a78c69f-d5ee-4a0d-bb76-1e52aa97ac56
Polymarket: highest temperature in Hong Kong on October 13, 2026	https://polymarket.com/zh-hant/event/highest-temperature-in-hong-kong-on-october-13-2026